SE LAB 05

Static Code Analysis

11_scholarship_screener

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ISSUES TABLE

No.	Issue	Type	Line(s)	Description	Fix Approach
1	Use of eval()	Security — Medium	12	The eval() function evaluates a string as Python code and may execute unintended or malicious expressions.	Replace eval() with a safe formula evaluator that permits only approved operations.
2	Bare except	Error Handling — Low	37–38	The exception handler catches all exceptions and silently ignores them using pass.	Catch specific exceptions and handle them explicitly.
3	Unused math import	Code Quality	2	The math module is imported but never used.	Remove the unused import.
4	Unused sys import	Code Quality	3	The sys module is imported but never used.	Remove the unused import.
5	Mutable default argument	Potential Bug	14	The default argument logs=[] creates a shared list across function calls.	Use logs=None and initialize a new list inside the function.
6	Boolean comparison	Code Style	43	Comparing a Boolean value with True is unnecessary.	Replace results[r]["eligible"] == True with if results[r]["eligible"]:.
7	Missing input validation	Potential Bug / Security	11–19, 31–33	GPA, credits, income, and the custom formula are not adequately validated.	Validate input types, allowed ranges, and formula operations before processing.
8	File handling without a context manager	Code Quality / Reliability	26–28, 36	Files are opened without a context manager, which can make reliable resource cleanup harder.	Use with open(...) and specify an appropriate encoding.
9	Incorrect blank-line spacing	Code Style	Multiple lines	Flake8 reports E302 and E305 because functions do not have the required blank-line spacing.	Add the required blank lines between top-level functions and after function definitions.
10	Old-style string formatting	Code Style	21	The log message uses the older % string-formatting method.	Replace it with an f-string, such as f"{student_id} scored {score}".
11	Non-standard function naming	Code Style	40	printShortlist() does not follow the conventional snake_case naming style.	Rename it to print_shortlist() and update its function call.